Reverse Engineering

Reverse engineering is the process of obtaining a 3D CAD model of a given physical part, typically by means of measurement or capturing points with 3D scanning methods. The models can then be used for rapid prototyping, simulations, animations, etc.

Generally, RE's working principle is capturing features to be used, unlike forward engineering where features are captured from guidelines. Hence, lesser skill and time is required, however a more adaptive approach is needed to empathise with the designer's intent.

Thus, RE provides a platform for designing new products, reducing production time while following the client's needs.

RE is useful for manufacturing, industrial design, medical implants and jewelry design.

Note that RE is not for copying, but as an competitor analysis method for us to improve on its designs or capture ideas to be used in our designs.

RE can be done by either manual metrology methods, or by 3D scanning.

Generally, in the case of scanners, the process involves

- scanning the part
- processing the point data from scanning
- Applying the point data, for surface models, inspection with CAD model, or triangulating into STL file.
- Afterwards, the resulting analysis from the scanned model can be used for prototyping/redesigning parts

3D Scanning Software

- The scanning process generally has 5 stages:
- Acquisition
  - the scanner produces a specific light pattern on the object to collect point data, first in a rough acquisition to obtain overall sizing, then multiple scans afterwards to finely capture details.
- Alignment

when multiple scans are done, they need to aligned to have different points correspond to the same feature, giving more definition in the point cloud. global or manual alignment may be used

Mesh Generation

the point cloud is triangulated to form a triangular mesh

Post Processing

the raw model may have some errors that need to be processed to be fixed. these include:

making manifolds; a continuous surface where an edge is between two faces only

Removing intersections, so that topological issues are managed

Filling holes; by detecting missing portions of the mesh and patching the hole to ensure a closed mesh.

Simplification

after the model has been fixed, it needs to be simplified for better performance and easier application. this includes smoothing surfaces to reduce inherent noise, and decimation, to reduce the point count for coplanar faces, which improves performance.

Typically, specific RE software is used for this, for importing point data, followed by reducing noise and number of points afterwards, by taking advantage of multiple scans.

3D Scanning Hardware

there are 3 main methods to physically scan parts for RE, contact, non contact, and destructive

Contact Scanners

contact scanners general use probes to measure coordinates, which can provide more accuracy at the expense of being slightly slower and being unable to scan softer materials.

the three main methods to make contact scanners are

Hard/Manual Probes

Slow, but simple and cheap.

Works by allowing probe to settle on surface, before coordinates are recorded by CMM.

Accuracy and reliability is heavily dependent on operator.

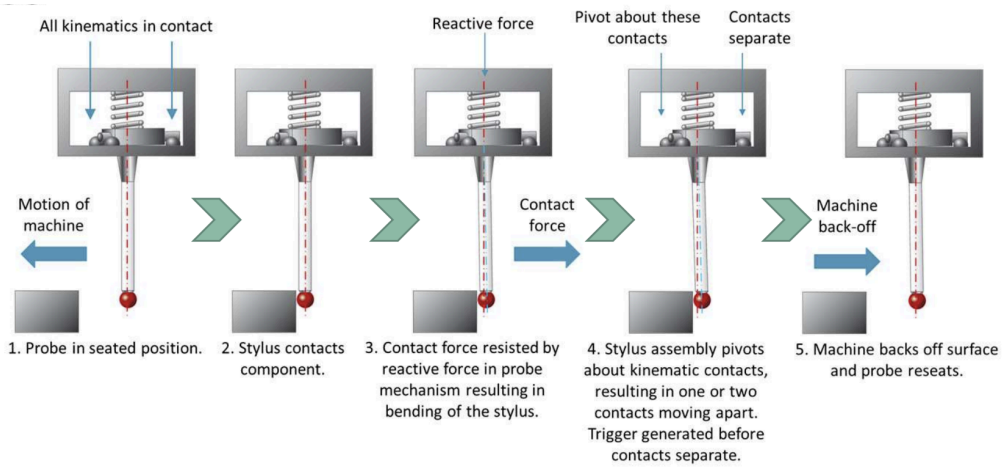
Hard Probe[2]

nexonar Hard Probe[1]

Point to Point sensing with touch trigger probes

probe does not need to settle >> electronic signal produced at contact which causes beep or light flash, making it rapid and reliable, eliminating variability due to operator.

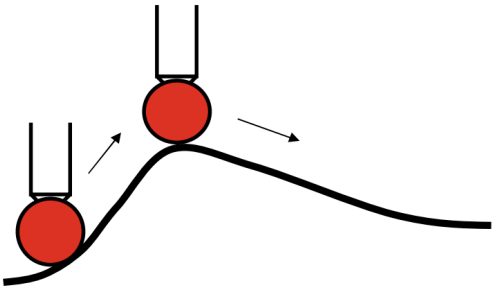
either uses piezo based sensors (eliminates stylus bending) or strain gauge sensors (triggered by constant force), which both cause higher accuracy.



Analog Sensing with Scanning Probes

for scanning curved surfaces and contours (crankshafts, cams, turbine blades, automobile bodies), hence requiring analog data input. (10 to 50x more data)

probe remains in contact with whole surface



In general, these methods have high accuracy, low cost, can measure deep recesses, and work with any solid object regardless of colour/transparency

However, data acquisition is slower and does not work with soft objects due to distortion

Non Contact Scanners

non contact scanners use optical methods like lasers, which works for soft materials but does not work with reflective or transparent materials.

The general working principle is that cross sections and point clouds are collected by transmitting energy onto the object (light, sound or magnetic fields), and the reflected energy is captured, and interpreted into geometrical data via triangulation, time of flight, and other algorithms.

There are 9 methods in total for this class of scanning:

Triangulation

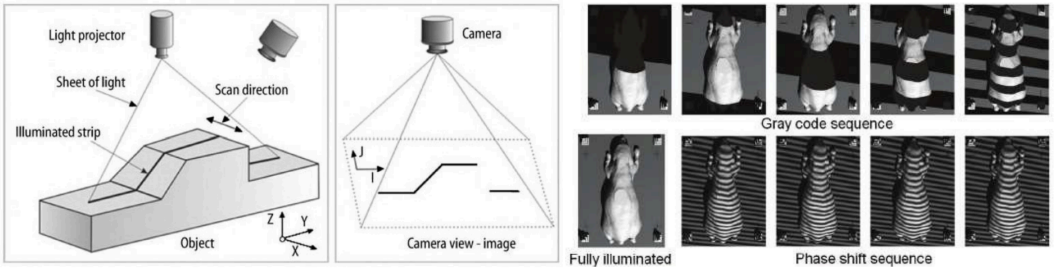
The most common method in laser scanners, utilising the location and angles of light sources and photosensitive devices to calculate coordinates, which can be triangulated to find surfaces.

Time of flight/Ranging Systems

Structured-Light & Stereoscopy

A known light pattern is projected onto the object at a certain angle and the image of the projected pattern on the object is captured,

which allows image analysis via triangulation work backwards on what surface causes the different projection.

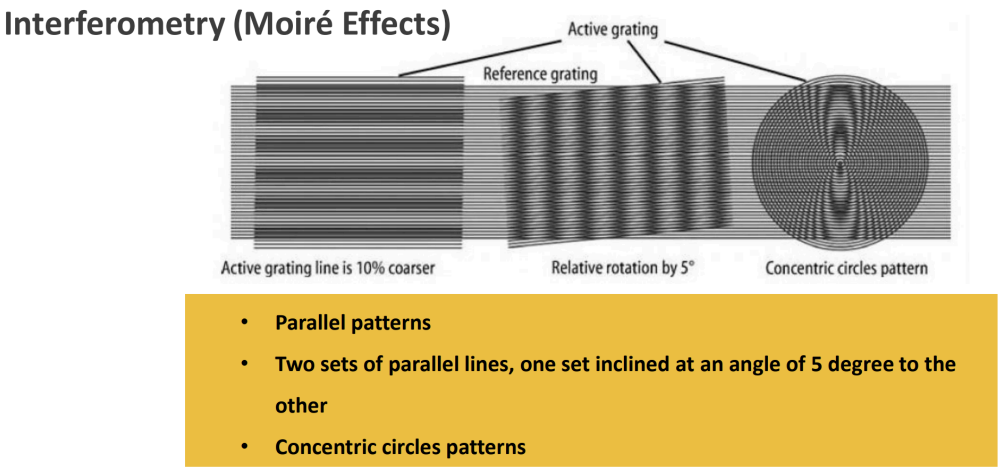


this method has a high data acquisition rate, while also capturing color and texture if needed, without needing lasers.

Typically, the light pattern used can be either a single stripe, multiple stripes, multiple dots, or a light grid.

One specific type of structured light method is interferometry:

- structured light patterns are shined onto surface to produce moire effect shadows.
- contours produced by shadows are captured and analyzed to find distance between lines, which is proportional to the height at the point.
- higher accuracy, best suited for smaller objects, struggles with larger objects due to larger range-accuracy trade off



Lasers

Image Analysis

Synchronised Scanning

Computer Tomography (CT Scanning)

Magnetic Resonance Imaging (MRIs)

Acoustic Methods

Generally, these methods are fast at large scale, have good accuracy and resolution, can detect colours, textures, and fine details, and works with flexible/soft/deformable solids, however struggles with transparent or reflective solids.

Destructive Methods

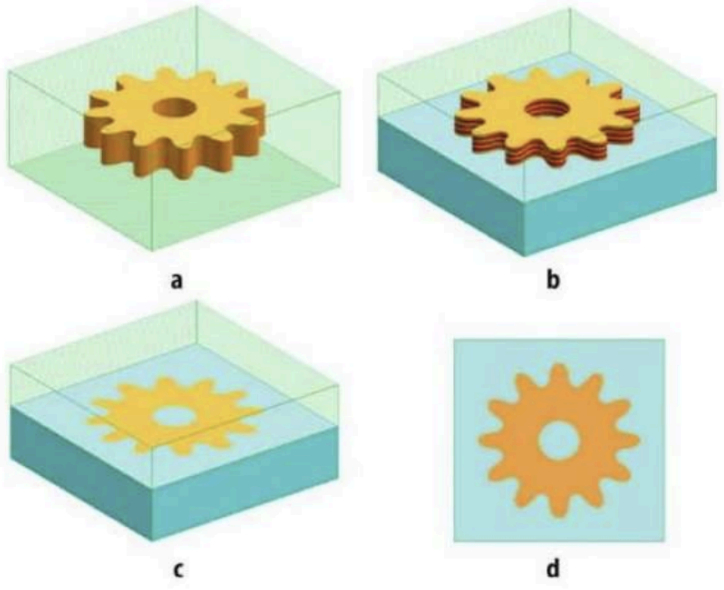
The main method for this is cross-section scanning (CSS).

Useful for small but complex objects, where internal & external features need to be scanned.

The method involves removing material layer by layer and capturing the cross sectional image with a CCD camera.

The images have edge detection algorithms run to acquire point data at the edges to form the point cloud.

To cut out consistent cross sections, the part is first embedded in a color-contrasting matrix and left to cure, before being machined as a block which is easier to do instead of the original shape of the part:



Despite being a destructive method, it is fast, repeatable and accurate to +-0.0127mm, can achieve high resolution with thin layers of 0.0127 to

0.254mm, and is versatile to any machinable material; wood, steel, plastics, etc.